

Exercise 5.8

$$\frac{1}{F} = \frac{1}{2E_a 2E_b v_{ab}}$$

$$v_{ab} = |V_a - V_b|$$

Assumption $V_a \parallel V_b \Rightarrow$ We can use centre of mass frame so that $\vec{p}_a = -\vec{p}_b$

$$\Rightarrow p_a = (E_a, 0, 0, |\vec{p}_a|), \quad p_b = (E_b, 0, 0, |\vec{p}_a|) \quad \left\| \begin{array}{l} \text{we use } E = \gamma m \text{ \& } \\ \vec{p} = \gamma m \vec{v} \\ \Rightarrow \vec{v} = \frac{\vec{p}}{E} \end{array} \right.$$

$$v_a = \frac{|\vec{p}_a|}{E_a}$$

$$\Rightarrow |v_a - v_b| = \left| \frac{|\vec{p}_a|}{E_a} - \frac{|\vec{p}_a|}{E_b} \right| = \frac{|\vec{p}_a| E_b - |\vec{p}_a| E_a}{E_a E_b} = v_{ab}$$

$$\Rightarrow F = 2E_a 2E_b v_{ab} = 4 |E_b - E_a| |\vec{p}_a|$$

$$E_a^2 = |\vec{p}_a|^2 + m_a^2, \quad E_b^2 = |\vec{p}_a|^2 + m_b^2$$

Let's assume for now that $m_a > m_b \Leftrightarrow E_a > E_b$ since for the opposite case we get the valid result by swapping the place of the indices.

$$\Rightarrow F = 4 (E_a - E_b) |\vec{p}_a| = 4 (E_a - E_b) \sqrt{E_a^2 - m_a^2}$$

$$= 4 \sqrt{(E_a^2 - m_a^2)(E_a - E_b)^2}$$

$$= 4 \sqrt{|\vec{p}_a|^2 (E_a^2 + E_b^2 - 2E_a E_b)}$$

$$= 4 \sqrt{E_a^2 |\vec{p}_a|^2 + E_b^2 |\vec{p}_a|^2 - 2E_a E_b |\vec{p}_a|^2}$$

$$= 4 \sqrt{E_a^2 E_b^2 + |\vec{p}_a|^4 - 2E_a E_b |\vec{p}_a|^2 - m_a^2 m_b^2}$$

$$= 4 \sqrt{(E_a E_b - |\vec{p}_a|^2)^2 - m_a^2 m_b^2}$$

$$\begin{aligned} m_a^2 m_b^2 &= (E_a^2 - |\vec{p}_a|^2)(E_b^2 - |\vec{p}_a|^2) \\ &= E_a^2 E_b^2 - E_a^2 |\vec{p}_a|^2 - E_b^2 |\vec{p}_a|^2 + |\vec{p}_a|^4 \\ \Rightarrow E_a^2 |\vec{p}_a|^2 + E_b^2 |\vec{p}_a|^2 &= -m_a^2 m_b^2 + |\vec{p}_a|^4 + E_a^2 E_b^2 \end{aligned}$$

$$p_a \cdot p_b = E_a E_b - |\vec{p}_a|^2$$

$$F = 4 \sqrt{(p_a \cdot p_b)^2 - m_a^2 m_b^2}$$

We can see the result is same under change of $a \leftrightarrow b$ so it works for both cases of $E_a > E_b$ & $E_b > E_a$

$$\lambda(s) = (s - m_a^2 - m_b^2)^2 - 4m_a^2 m_b^2 \quad \parallel \text{ with } s = (p_a + p_b)^2 = m_a^2 + m_b^2 + 2p_a p_b$$

$$= s^2 - 2s[m_a^2 + m_b^2] + [m_a^2 + m_b^2]^2 - 4m_a^2 m_b^2$$

$$= \underbrace{[m_a^2 + m_b^2]^2}_{\text{cancel}} + \underbrace{4p_a p_b [m_a^2 + m_b^2]}_{\text{cancel}} + 4(p_a p_b)^2 - 2(\underbrace{[m_a^2 + m_b^2]}_{\text{cancel}} + 2p_a p_b)[m_a^2 + m_b^2] + \underbrace{[m_a^2 + m_b^2]^2}_{\text{cancel}} - 4m_a^2 m_b^2$$

$$\Rightarrow \lambda(s, m_a^2, m_b^2) = 4 (p_a p_b)^2 - 4 m_a^2 m_b^2 = 4 [(p_a p_b)^2 - m_a^2 m_b^2]$$

$$\Rightarrow F = 4 \sqrt{(p_a p_b)^2 - m_a^2 m_b^2} = 4 \sqrt{\frac{1}{4} \lambda(s, m_a^2, m_b^2)}$$

$$\Rightarrow F = 4 \sqrt{(p_a p_b)^2 - m_a^2 m_b^2} = 2 \sqrt{\lambda(s, m_a^2, m_b^2)} \quad \square$$

The significance of the result is that we can write the flux in terms of Lorentz invariant quantities when the particles are colliding such that $v_a \parallel v_b$.

Exercise 5.9

The n-particle phase space is

$$dPS_n = \left(\prod_{i=1}^n \frac{d^3 p_i}{(2\pi)^3 2E_i} \right) (2\pi)^4 \delta^{(4)}(p_a + p_b - \sum_i p_i)$$

In the C.M. - Frame we have $p_a + p_b = (2E, 0, 0, 0)$

Also $\vec{p}_a = -\vec{p}_b$, so we get

(This is centre of mass energy E_{cm})

$$\text{get } \delta^{(4)}(p_a + p_b - \sum_i p_i) = \delta(E_{cm} - \sum_i E_i) \delta^{(3)}(-\sum_i \vec{p}_i)$$

In n=2 situation we get

$$\int dPS_2 = \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6 4E_1 E_2} (2\pi)^4 \delta(E_{cm} - (E_1 + E_2)) \delta^{(3)}(-(\vec{p}_1 + \vec{p}_2))$$

Integrating over p_2 we get $(E_1 = E_2, \vec{p}_1 = -\vec{p}_2)$

$$\int dPS_2 = \int \frac{d^3 p_1}{16\pi^2 E_1^2} \delta(E_{cm} - 2E_1) \quad \parallel \quad E_1 = \sqrt{\vec{p}_1^2 + m_1^2}$$

$$\int dPS_2 = \int \frac{d^3 p_1}{16\pi^2} \frac{1}{|\vec{p}_1|^2 + m_1^2} \delta(E_{cm} - 2\sqrt{|\vec{p}_1|^2 + m_1^2})$$

We move on to spherical coordinates $d^3 p_1 = |p|^2 d\Omega dp$ with $|p| = |\vec{p}_1|$

$$\Rightarrow \int dPS_2 = \int d\Omega \int \frac{dp}{16\pi^2} \frac{|p|^2}{|p|^2 + m_1^2} \delta(E_{cm} - 2\sqrt{|p|^2 + m_1^2})$$

We use the identity for delta function:

$$\delta(g(x)) = \frac{\delta(x-x_0)}{|g'(x_0)|} \quad \text{where } g(x_0) = 0$$

As such

$$\delta(E_{cm} - 2\sqrt{|p|^2 + m_i^2}) = \frac{\delta(|p| - \sqrt{\frac{1}{4}E_{cm}^2 - m_i^2})}{\left| -\frac{1}{2} \cdot 2 \cdot \frac{2\sqrt{\frac{1}{4}E_{cm}^2 - m_i^2}}{\sqrt{\frac{E_{cm}^2}{4} + m_i^2}} \right|}$$

$$\delta(E_{cm} - 2\sqrt{|p|^2 + m_i^2}) = \frac{\delta(|p| - \sqrt{\frac{1}{4}E_{cm}^2 - m_i^2}) E_{cm}}{4\sqrt{\frac{E_{cm}^2}{4} - m_i^2}}$$

$$\Rightarrow \int dPS_2 = \int d\Omega \int \frac{d|p|}{16\pi^2} \frac{|p|^2 E_{cm}}{|p|^2 + m_i^2} \frac{\delta(|p| - \sqrt{\frac{1}{4}E_{cm}^2 - m_i^2})}{4\sqrt{\frac{E_{cm}^2}{4} - m_i^2}}$$

$$\int dPS_2 = \int d\Omega \frac{E_{cm}}{16\pi^2} \frac{\sqrt{\frac{E_{cm}^2}{4} - m_i^2}^2}{E_{cm}^2/4 + m_i^2 - m_i^2} \frac{1}{4\sqrt{\frac{E_{cm}^2}{4} - m_i^2}} = |p|$$

$$dPS_2 = \int d\Omega \frac{E_{cm}}{16\pi^2} \frac{|p|}{E_{cm}^2}$$

$$\Rightarrow \underline{dPS_2 = \int d\Omega \frac{1}{16\pi^2} \frac{|p|}{E_{cm}}}$$

The cross section is defined as $\sigma_{ab \rightarrow 12} = \frac{1}{F} \int dPS_2 |M(p_a, p_b \rightarrow p_1, p_2)|^2$

The matrix element doesn't depend on p_i so we can use the previous result. In CM-frame: This is because it only depends on fundamental

$$(\sigma_{ab \rightarrow 12})_{cm} = \frac{1}{F} \int d\Omega \frac{1}{16\pi^2} \frac{|p|}{E_{cm}} |M(p_a, p_b \rightarrow p_1, p_2)|^2$$

interactions & not the kinematics.

$$\Rightarrow \left(\frac{d\sigma}{d\Omega} \right)_{cm} = \frac{1}{F} \frac{|p|}{16\pi^2 E_{cm}} |M(p_a, p_b \rightarrow p_1, p_2)|^2$$

Exercise 5.10

Suppose there is interaction of form $\frac{\lambda}{3!} \phi^3$

Calculate the amplitude for two-to-two scattering in tree level approximation

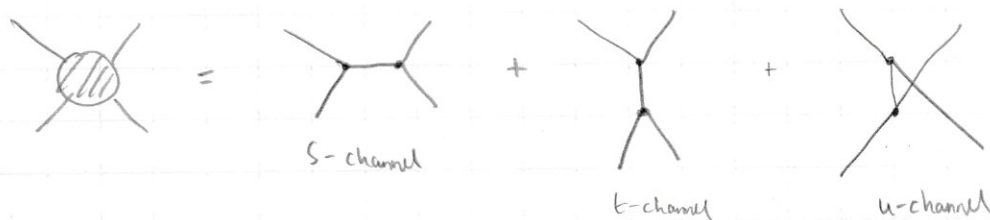
We start by listing the Feynman rules:

For the vertex  we have $-i\lambda$

for the propagator  we have $-\frac{i}{p^2 - m^2}$

and the external legs gives  just 1

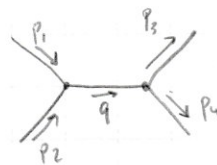
In 2-2 scattering (of identical ^{scalar} particles) we get the Matrix elements to be sum of s-, t- and u-channel matrix elements



Using the Feynman rules we get

$$i\mathcal{M}_s = 1 \cdot (-i\lambda) \cdot 1 \cdot \frac{-i}{q^2 - m^2} \cdot 1 \cdot (-i\lambda) \cdot 1$$

$$= \frac{\lambda^2 i}{s - m^2}$$

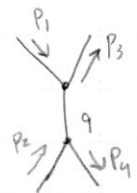


$$q^2 = (p_1 + p_2)^2 = s$$

$$i\mathcal{M}_t = 1 \cdot (-i\lambda) \cdot 1 \cdot \frac{-i}{q^2 - m^2} \cdot 1 \cdot (-i\lambda) \cdot 1$$

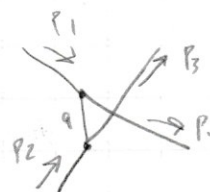
$$= \frac{\lambda^2 i}{t - m^2}$$

$$\text{with } q^2 = (p_1 - p_3)^2 = t$$



$$i\mathcal{M}_u = 1 \cdot (-i\lambda) \cdot 1 \cdot \frac{-i}{q^2 - m^2} \cdot 1 \cdot (-i\lambda) \cdot 1$$

$$= \frac{\lambda^2 i}{u - m^2}$$



$$q^2 = (p_1 - p_4)^2 = u$$

$$\Rightarrow |\mathcal{M}|^2 = |\mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u|^2 = \lambda^4 \left[\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right]^2$$

We can compute the differential cross-section using result of previous exercise.

In centre of mass frame we have equation of differential cross section:

$$\left(\frac{d\sigma}{d\Omega}\right)_{cm} = \frac{1}{F} \frac{|p_1|}{16\pi^2 E_{cm}} \underbrace{|\mathcal{M}(p_1, p_2 \rightarrow p_3, p_4)|^2}_{=|\mathcal{M}|^2 \text{ that we just calculated}}$$

$$E_{cm} = p_1 + p_2 = \sqrt{s}$$

$$\Rightarrow \left(\frac{d\sigma}{d\Omega}\right)_{cm} = \frac{\lambda^4}{F} \frac{|p_1|}{16\pi^2 \sqrt{s}} \left[\frac{1}{s-m^2} + \frac{1}{t-m^2} + \frac{1}{u-m^2} \right]^2$$